

Hot Hand, Real or Believed?

A behavioural analysis of setting decisions in professional volleyball

FINAL RESULTS

Three questions for one dataset

RQ1

Real hot hand

Does success predict a hitter's success on her next attack?

RQ2

Believed hot hand

Does the setter choose the same hitter again after a successful attack?

RQ3

Competitive pressure

Does a critical score situation change attack error rate or the attacking situation the team is able to create?

63 matches, two teams, mixed-effects models

63

matches parsed
with pydatavolley

8,312

own-team
attacks analysed

23

attackers
in the dataset

2

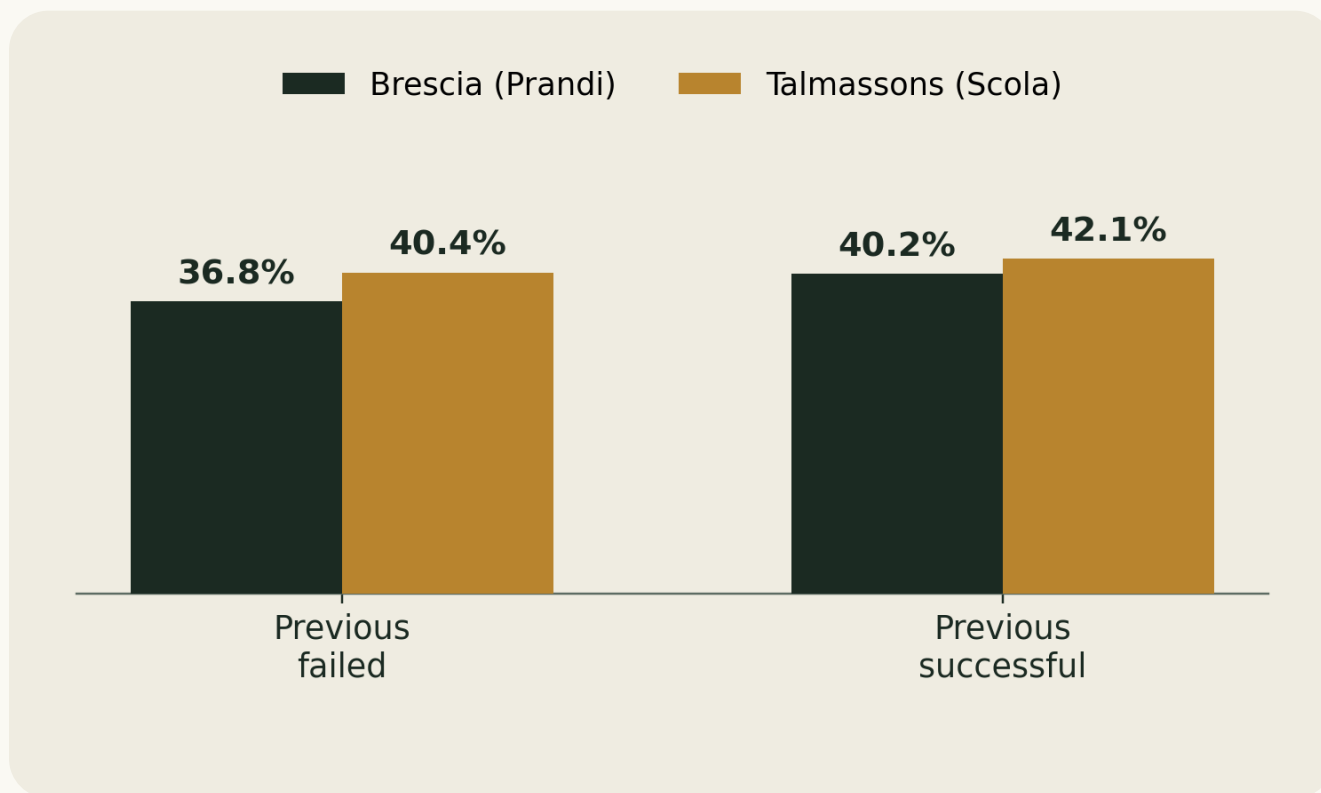
setters:
Prandi & Scola

Raw Data Volley (.dvw) files parsed with pydatavolley and checked through additional consistency tests

Logistic mixed-effects models with random intercepts for player and match were used to account for repeated observations and baseline differences.

Raw comparison: an apparent Brescia effect

Attack success rate, by outcome of the previous attempt (raw rates)



Brescia: success rate is 3.4 percentage points higher after a previous success.

Talmassons: the difference is smaller, +1.7 percentage points.

Looking only at raw rates, Brescia appears to show a possible hot-hand pattern

The mixed model: no significant hot-hand effect

After accounting for differences in baseline success between players and matches, the hot-hand effect is not statistically significant for either team.

BRESCIA (PRANDI)

$$\beta = +0.085$$

SE 0.064 · not significant

A positive but small effect: no reliable evidence of a real hot hand.

TALMASSONS (SCOLA)

$$\beta = +0.031$$

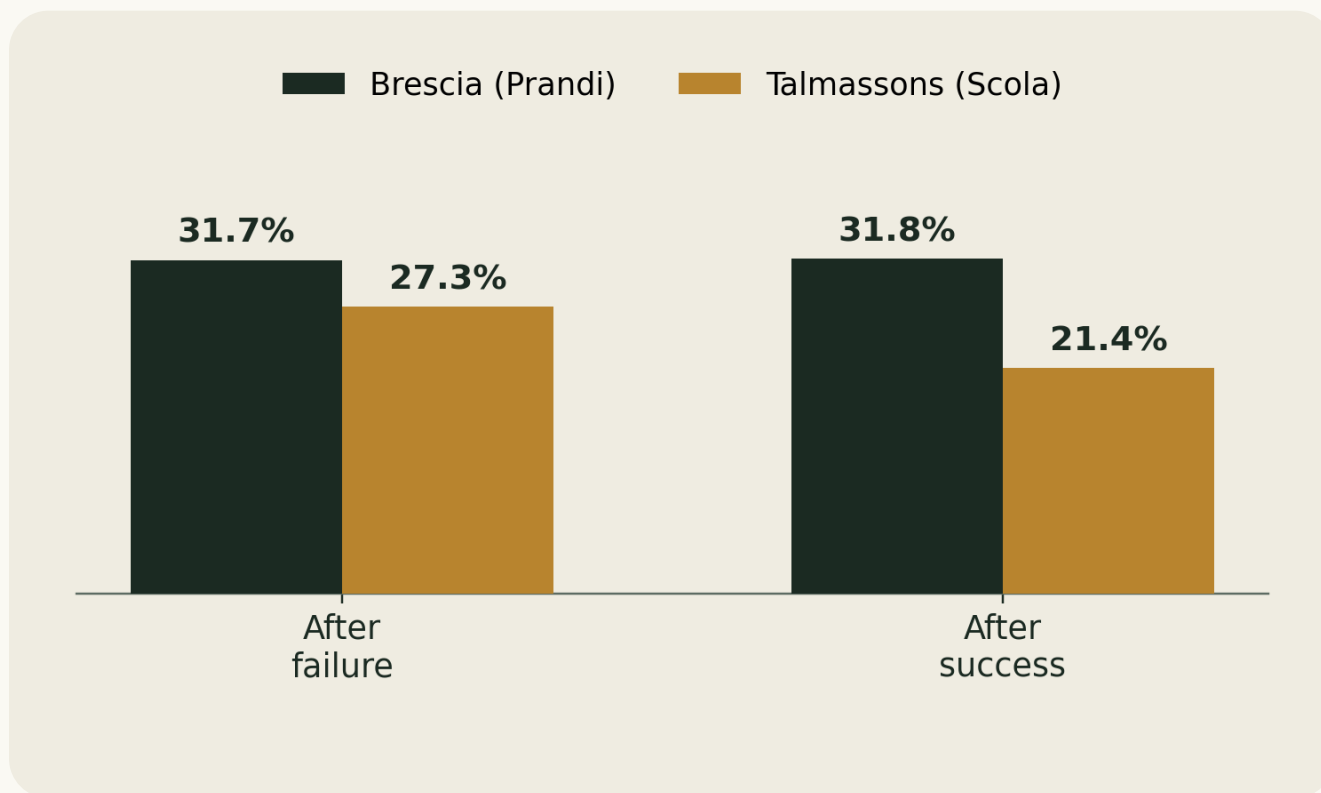
SE 0.071 · not significant

The effect is even closer to zero than for Brescia.

The raw Brescia pattern was therefore not enough to support a real hot-hand effect.

Raw comparison: different setting patterns

Rate of choosing the same hitter again, by outcome of her attack (raw rates)



Brescia: almost no difference after a successful attack (+0.1 percentage points).

Talmassons: the same hitter is selected less often after a success (−6.0 percentage points).

The descriptive pattern is therefore much stronger for Talmassons than for Brescia.

The mixed model: the Talmassons pattern remains

Unlike the RQ1 result, the Talmassons pattern remains significant after accounting for differences between hitters and matches.

BRESCIA (PRANDI)

$$\beta = +0.047$$

SE 0.067 · not significant

No evidence that the setter changes her choice based on the previous outcome.

TALMASSONS (SCOLA)

$$\beta = -0.295***$$

SE 0.082 · significant ($p < .001$)

A successful attack is associated with a lower probability of selecting the same hitter again.

One structural explanation still has to be ruled out before reading this as a behavioural result.

Is this a rotation artefact?

A team rotates when it wins a rally while receiving. As a result, previous success and the attacking options available on the next action can be related through the rules on the game

THE MECHANISM

After a successful attack: the team rotates ~70% of the time.

After an unsuccessful one: only ~15%.

And when it rotates, the same hitter is re-targeted clearly less often.

Effect of success	Model A	Model B
Talmassons (Scola)	-0.283***	-0.142 n.s.
Brescia (Prandi)	+0.043 n.s.	+0.246***

Both models on the same sample ($n = 8,182$). Model B adds a control for whether a rotation occurred.

The interpretation changes

Scola's initial negative pattern is no longer significant, while a positive allocation pattern emerges for Prandi

RQ1 vs. RQ2: main results

	Brescia (Prandi)	Talmassons (Scola)	Pattern
RQ1 — Real hot hand	$\beta = +0.085$, n.s.	$\beta = +0.031$, n.s.	No significant hot-hand effect
RQ2 — Believed hot hand (uncontrolled)	$\beta = +0.043$, n.s.	$\beta = -0.283^{***}$	Negative pattern for Talmassons before rotation control
RQ2 — Believed hot hand (rotation controlled)	$\beta = +0.246^{***}$	$\beta = -0.142$, n.s.	Positive allocation pattern for Brescia

n.s. = not significant. Significance: * $p < .10$; ** $p < .05$; *** $p < .01$

No significant hot hand effect in either team , but Brescia's setter si more likely to return to a successful attacker after controlling for rotation

Does pressure change how the team attacks?

Critical phase: either team has reached 20 points in sets 1–4, or the match is in the fifth set. Two outcomes are tested: the attack error rate, and how often the team uses a high ball

Choking under pressure

Performance may decrease in high-pressure situations.

Clutch performance

Performance may instead improve when pressure is higher.

27.3% of attacks occur in critical phases

8,312

Total attacks
analysed

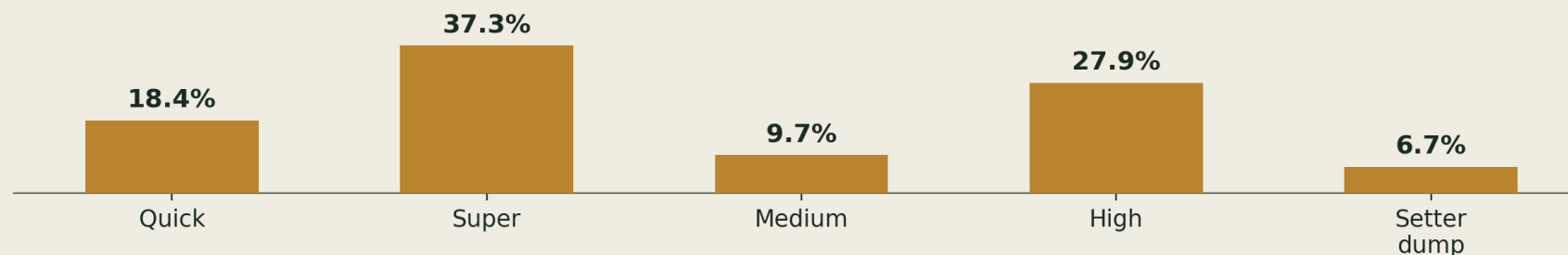
2,273

Attacks in
critical phases

27.3%

Of all
attacks

ATTACK TEMPO DISTRIBUTION (WHOLE SEASON)

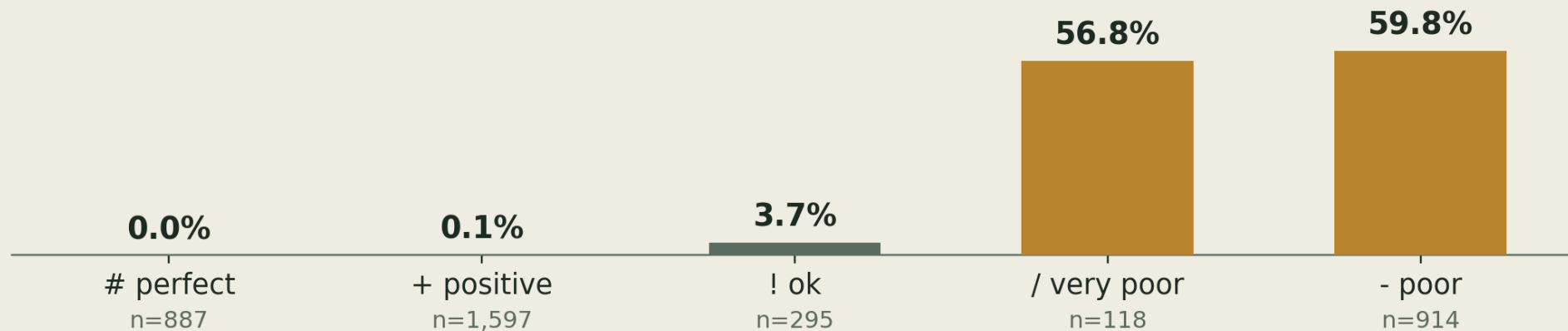


High balls (27.9% of all attacks) are the second outcome tested in RQ3 — but they need interpreting first.

High ball use depends on the build-up

High balls should not simply be interpreted as a more conservative attacking option. Faster attacks require a good reception or defensive action, while a poor first contact makes a high ball more likely

SHARE OF ATTACKS PLAYED ON A HIGH BALL, BY QUALITY OF THE PRECEDING RECEPTION



High-ball use can therefore also be considered an indirect indicator of the quality of the build-up before the attack

Brescia changes its attacking pattern in critical phases

ATTACK ERROR — CRITICAL VS. NORMAL

Talmassons: $\beta = -0.046$, n.s. — no significant difference.

Brescia: $\beta = -0.358^{**}$ — significantly fewer errors.

IGH-BALL USE — CRITICAL VS. NORMAL

Talmassons: $\beta = -0.112$, n.s. — no significant difference.

Brescia: $\beta = -0.190$ — significantly lower high-ball use**

Talmassons remains stable, while Brescia shows significant changes in both outcomes.

No evidence of choking: a possible clutch-like pattern

TALMASSONS

No significant change in either outcome during critical phases

BRESCIA

Significantly fewer attack errors during critical phases.

High-ball use is also significantly lower. Given its relationship with the preceding actions, this is consistent with Brescia reaching the attack in better conditions

Overall, there is no evidence of worse attacking performance under pressure. Brescia shows a possible clutch-like pattern, although the mechanism behind it cannot be established from these models alone

Separating the phenomenon from the belief

RQ1 — Performance

The raw data suggested a possible hot-hand pattern for Brescia, but this was not significant in the mixed model: the difference may be partly explained by baseline differences between hitters

RQ2 — Setting decisions

Rotation had to be separated from choice. Once it is, Talmassons' apparent avoidance disappears, and Brescia's setter shows a significant tendency to return to a hitter who has just scored

The dissociation

Brescia's hitters give show no significant hot hand effect, while the setter is more likely to return to a successful attacker after controlling for rotation

Whether this is a deliberate strategy or an unnoticed bias cannot be settled from scouting data alone.

What this project does not yet resolve

Rotation

Rotation is now controlled, but only as a binary marker. A finer version would use the hitter's actual position, separating front-row from back-row situations.

Success definition

Success is defined as a direct point (#). Including positive attacks could be tested as a robustness check.

Data parsing

pydatavolley is an open-source parser, so its output required additional validation and consistency checks.

Critical phases

Only 27.3% of attacks are classified as critical, limiting the amount of data available for this condition.

Performance and setting behaviour show different patterns

RQ1 — Real hot hand

No significant hot-hand effect was found for either team after accounting for hitter and match differences.

RQ2 — Believed hot hand

Once rotation is controlled, Brescia's setter is significantly more likely to return to a hitter who has just scored; Talmassons' initial negative pattern is no longer significant after controlling for rotation.

RQ3 — Competitive pressure

No evidence of worse attacking performance under pressure. Brescia shows fewer errors and lower high-ball use during critical phases.

Performance patterns and behavioural responses to previous outcomes need to be analysed separately.

Data sources and analysis tools

pydatavolley

parsing of raw Data Volley (.dvw) scouting files

Python / NumPy

data processing, statistical analysis and mixed-effects models

Dataset

regular season, Coppa Italia and play-offs

GitHub repository

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Hot Hand, Real or Believed?

No significant real hot hand effect but a positive allocation pattern emerges for Brescia after controlling for rotation.

The results show why performance, setting decisions and structural constraints need to be considered separately.